

Product Specification

SPECIFICATION
FOR
APPROVAL

() Preliminary Specification
(●) Final Specification

Title	Customer Approval Sheet for Lenovo 6.95 WS
-------	--

Customer	Lenovo
Part Number	-

SUPPLIER	BOE
MODEL	TV070WSM-TL5
Version	P2

SIGNATURE	DATE
/	
/	
/	

Please return 1 copy for your confirmation with your signature and comments.

APPROVED BY	DATE
刘正道	2020/04/11
REVIEWED BY	

CONTENT LIST

4. 27. 2020

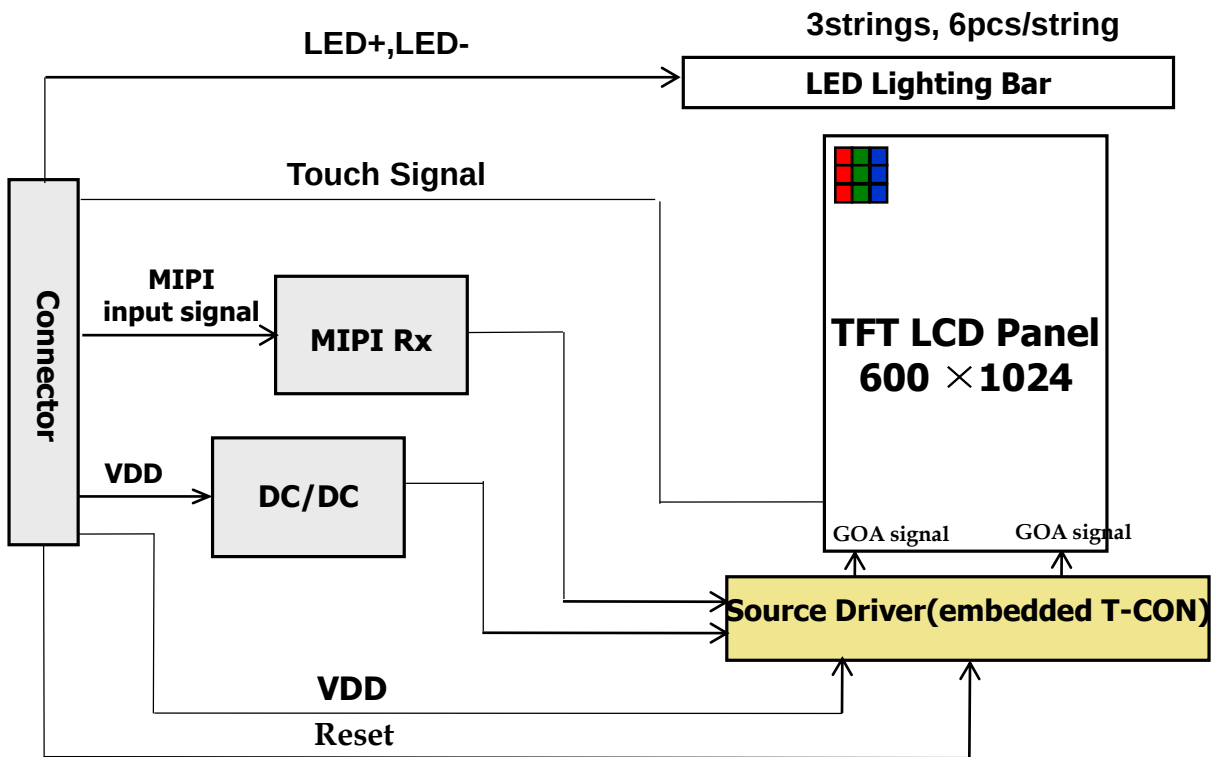
RECORD OF REVISIONS

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Product Specification

1-1. GENERAL DESCRIPTION

Block Diagram



Features

TV070WSM-TL5 is 6.95" 8bit color TFT-LCD (A-Si Thin Film Transistor Liquid Crystal Display) module composed of LCD panel, **MIPI** driver IC, control circuit, backlight and **Cover lens**. By applying 8 bit digital data, 600×RGB (3) ×1024, 16.7M-color images are displayed on the 6.95" diagonal screen

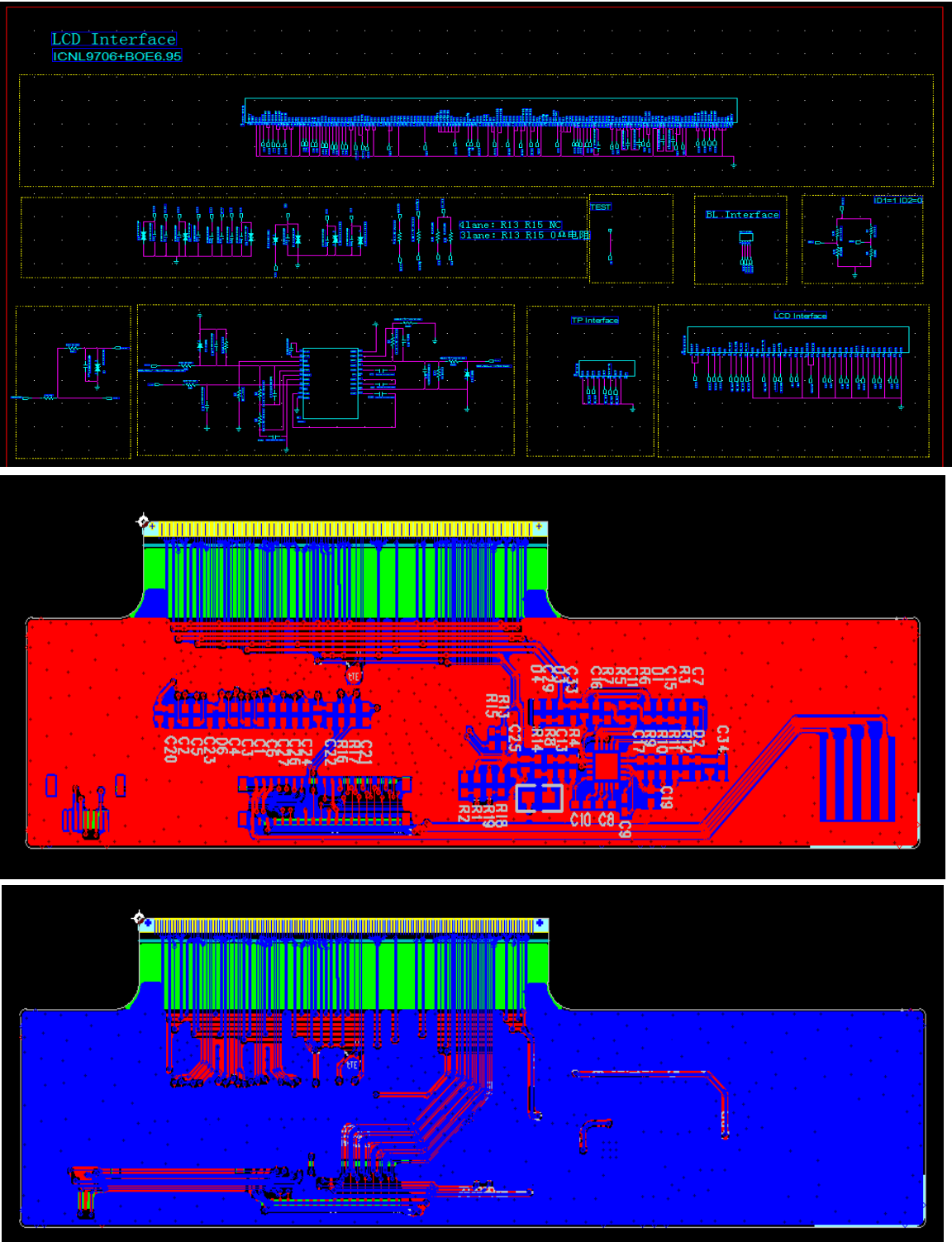
Product Specification

1-2. General Spec

No	Item	Specification	unit	Remark
1	Screen Size	6.95	inch	
2	Active Area	89.28*152.3712	mm	
3	Panel Size	93.28*159.82	mm	
4	Outline Dimension	97.74*171.14*3.3(Typ.)(TLCM)	mm	
5	Display Resolution	600*1024	pixel	
6	Pixel Pitch	49.6*148.8	um	
7	Touch Structure	Signal Layer On-cell	-	
8	Display Mode	Normally Black	-	
9	Display Color	16.7M	-	
10	Color Gamut	50	%	typ
11	Luminance	300 (TLCM)	nit	Min
12	Contrast Ratio	800:1	-	typ
13	Viewing Angle	85/85/85/85	°	CR>10(U/D/L/R)
14	Pol Surface Treatment	-	-	HC/AG
15	Weight	140(TLCM)	g	max
16	D-IC	ICN9706	-	
17	Inversion Method	Z-Inv		Dot/Column
18	LED Q'ty	18(3S*6P)	ea	String*Parallel
19	Power Consumption	1152mW@BLU Max 300mW @Logic Max.	mw	Backlight + Logic (RGB pattern)

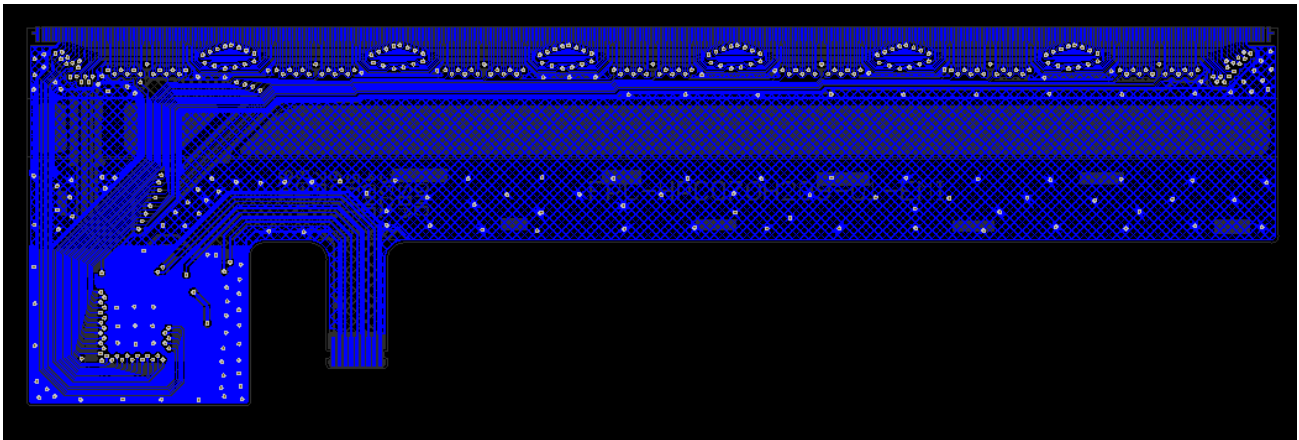
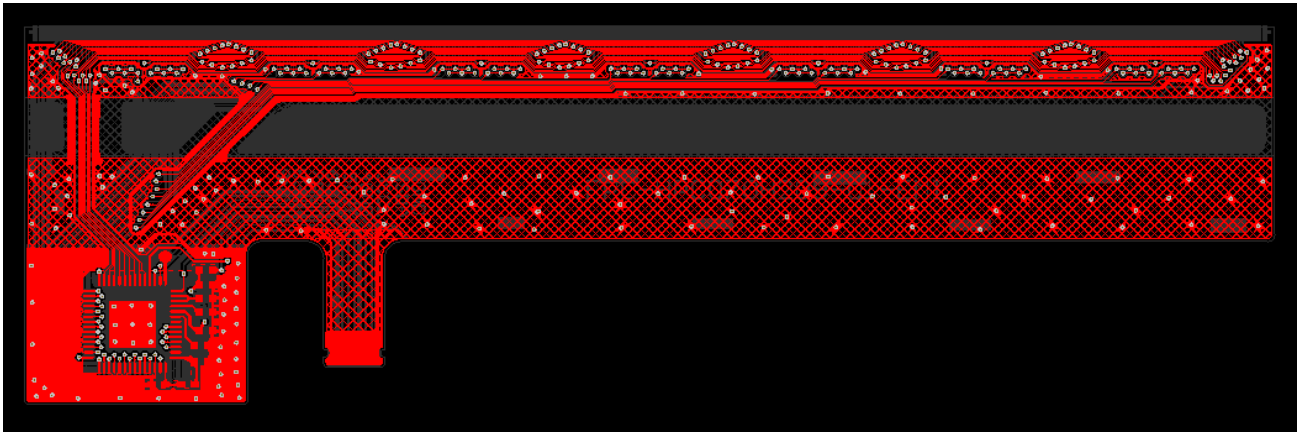
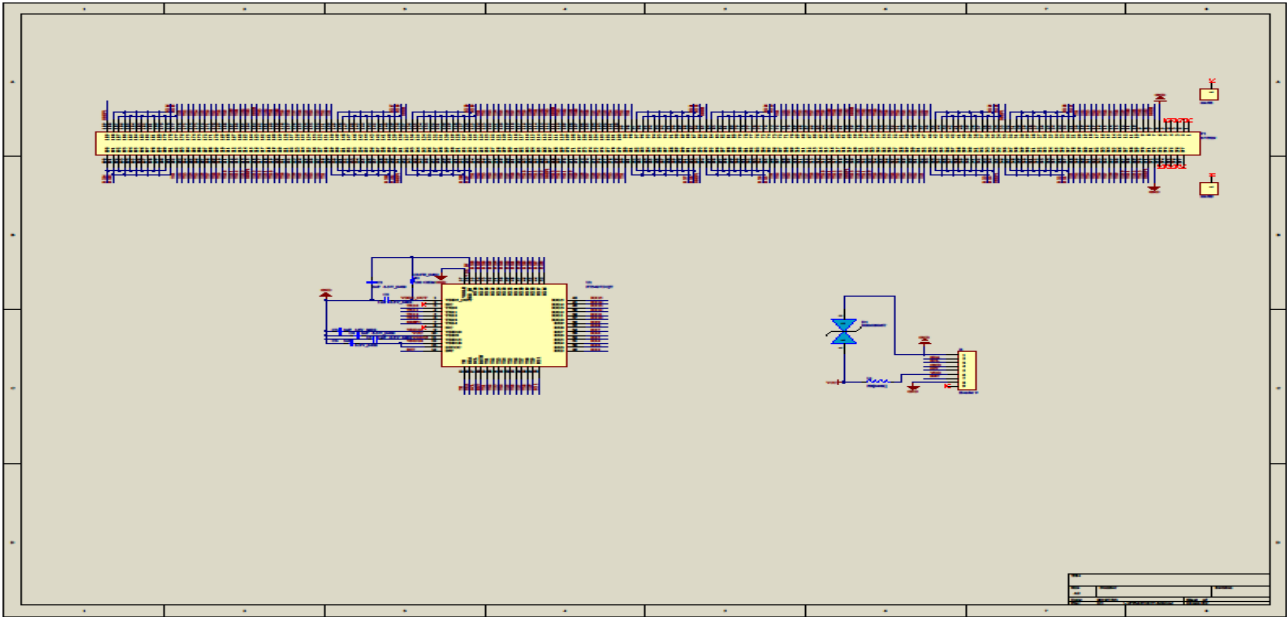
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1-3-1. LCM FPCA Gerber/Layout and Schematic Diagram



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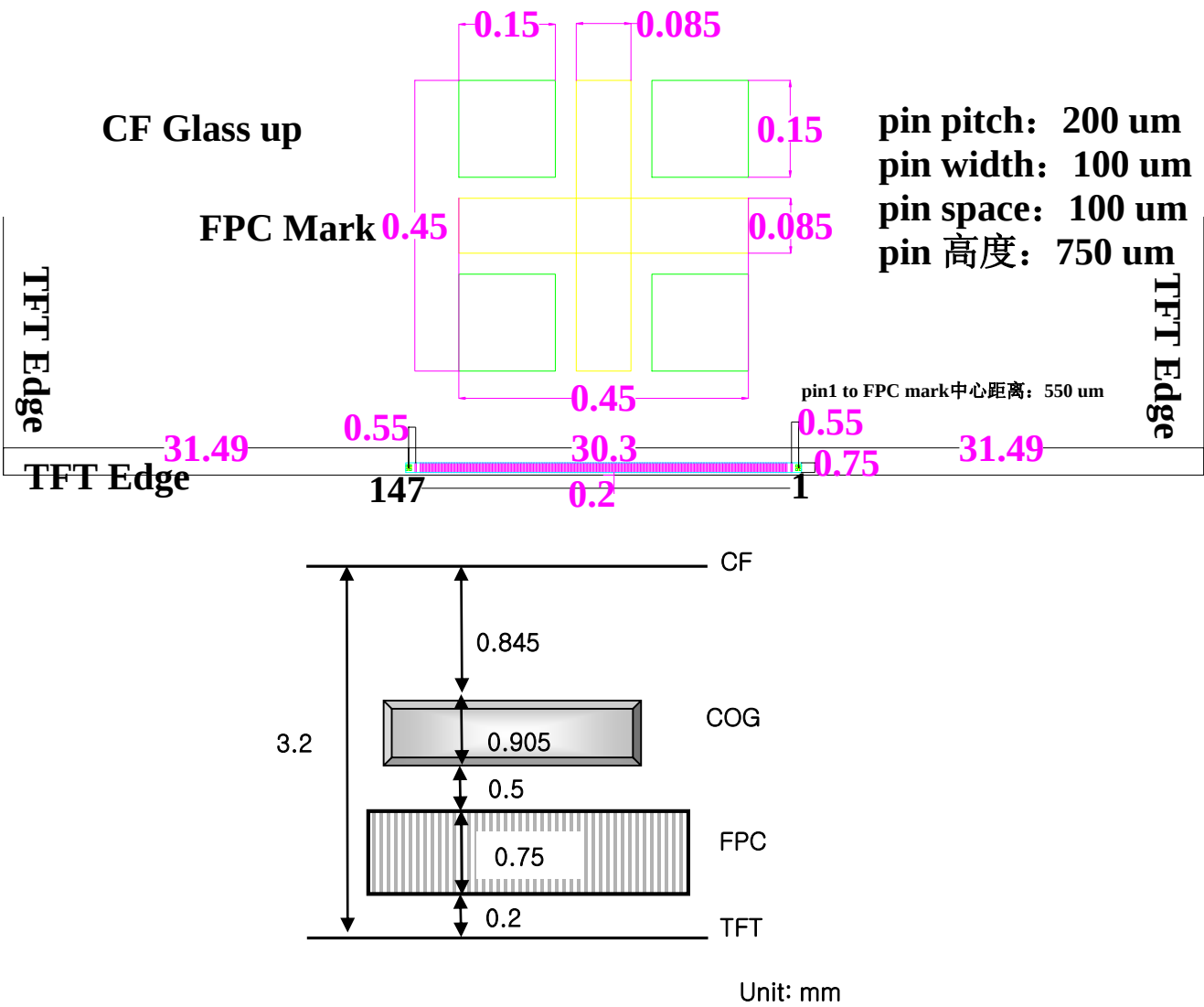
1-3-2. TP FPCA Gerber/Layout and Schematic Diagram



Product Specification

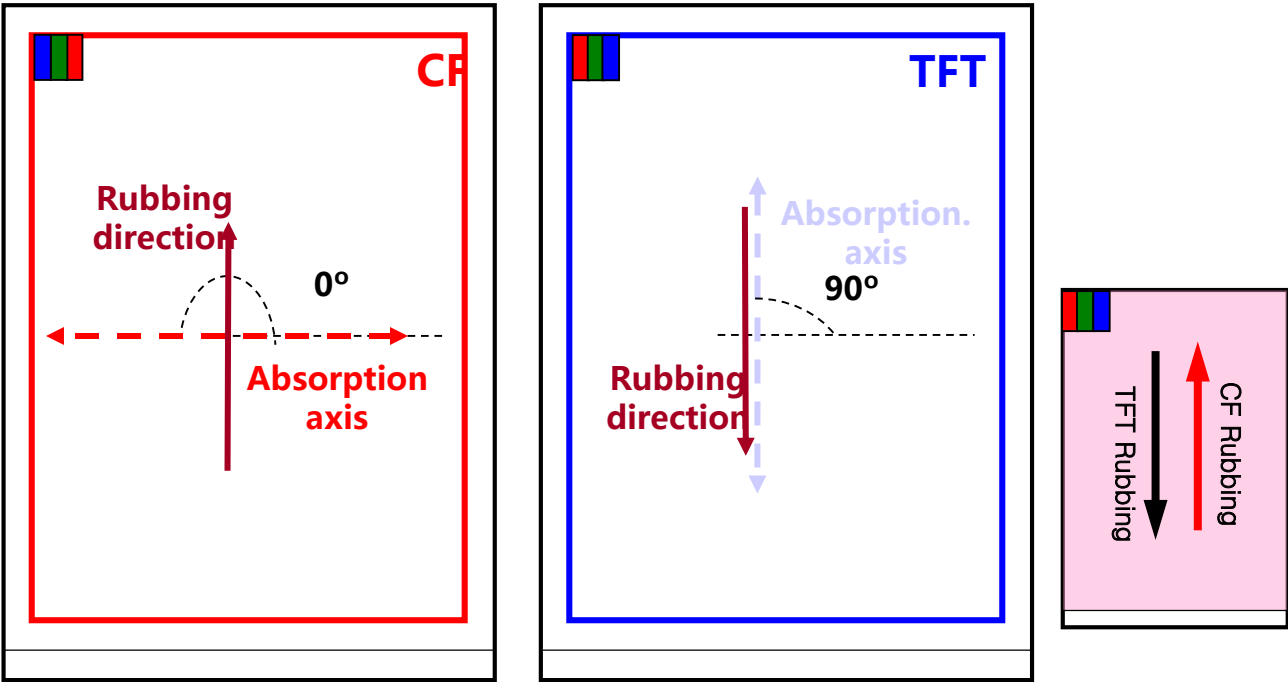
1-3-3. FPC Pin Assignment

Please pay attention that IC bump down(TFT glass up and C/F glass down)



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1-3-4. Pol General Spec



Item	C/F Pol	TFT Pol	Remark
Transmittance Axis	0°	90°	O-mode
Treatment	HC	Clear	-
Transmittance	43%	43%	-
Size	154.85*92.68	154.27*91.18	mm
Thickness	130	130	um

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2-1. Absolute Maximum Ratings

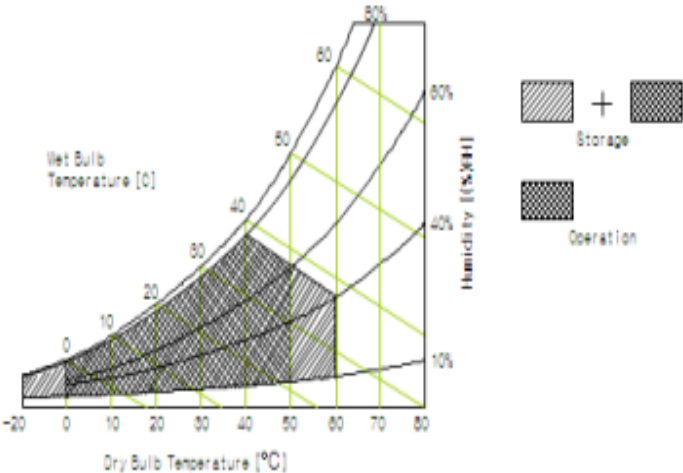
The following are maximum values which, if exceeded, may cause faulty operation or damage to the unit.

Table 2.1 Absolute Maximum Ratings

Parameter	Symbol	Values		Units	Notes
		Min	Max		
LCD Input Power Voltage	VDD	-0.3	3.6	V	T=25°C Note1&2
Touch Input Power Voltage	TP_VDDI	2.7	3.6	V	
LED Current	I _{LED}	-	20	mA	
Storage Humidity	Hstg	10%	90%	%RH	Note3
Storage Temperature	Tstg	-20°C	60°C	°C	

Note:

- 1. These range above is maximum value not the actual operating temperature . Actual Operating temperature is no more than 40°C and temperature refers to the LCM surface temperature ; Length of operation: No more than 8 hours per day, and no more than 4 hours of continuous use one time.
- 2. BOE is not responsible for product problems beyond the use conditions.
- 3. Temperature and relative humidity range are shown in the figure below. Wet bulb temperature should be 39 °C max. and no condensation of water.



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3-1. ELECTRICAL CHARACTERISTICS

Table 3.1 Electrical Characteristics Of TFT-LCD Module

Parameter		Symbol	Values			Units	Notes
			Min	Typ	Max		
LCD Input Voltage		VDD-Vss	2.7	3.3	3.6	V	If necessary
LED Input Current		I _{LED}	-	20	-	mA	
“H” Level Input Voltage		V _{IH}	0.7IOVCC	-	IOVCC	V	IOVCC=V _{DD}
“L” Level Input Voltage		V _{IL}	0	-	0.3IOVCC	V	
“H” Level Output Voltage		V _{OH}	0.8IOVCC	-	IOVCC	V	
“L” Level Output Voltage		V _{OL}	0	-	0.2IOVCC	V	
LCD Power Consumption	Normal	P _N	-	292	300	mW	1
	Deep standby	P _D	-	33	40	mW	
	BLU	P _B	-	1080	1152	mW	2

Notes:
(1) The specified current and power consumption are under the conditions at VDD = 3.3V, T = 25°C, and fv = 60 Hz, at Red (Green or Blue) pattern
(2) LED Backlight assumptions: 3.0 Vf, 20 mA,

3-2. Logic Power Consumption

Parameter	Symbol	Values		Units	Notes
		Typ	Max		
Normal Mode	I _{VDD}	90	-	mA	RGB pattern
Sleep Mode	I _{VDD}	10	-	mA	

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3-3. BACK LIGHT UNIT

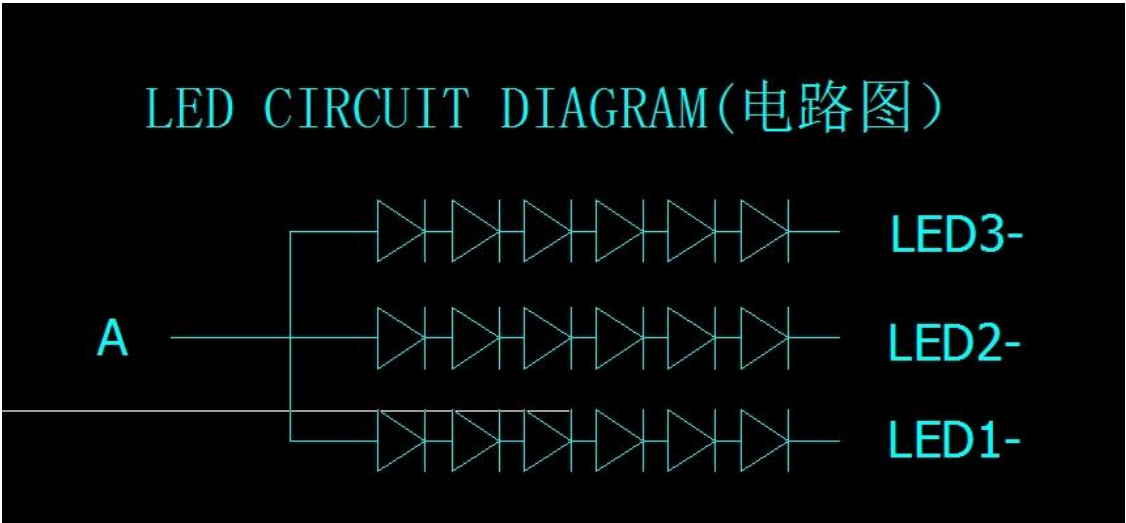
3-3-1 The edge-lighting type of back light unit consists of 18 LEDs which is connected in serial.

Table 2-3-1 Electrical Characteristics Of Back Light Unit

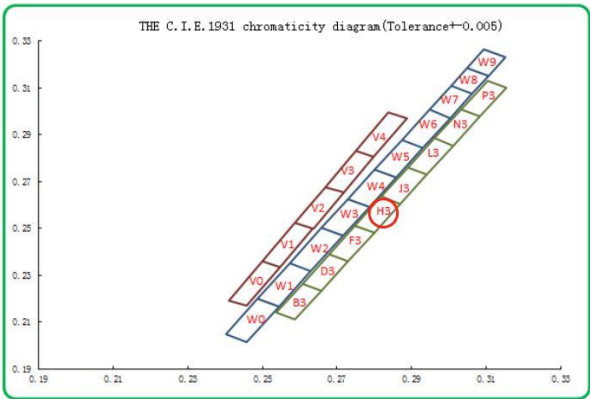
Parameter	Symbol	Values			Units	Notes
		Min	Typ.	Max		
LED Current	I _{LED}	-	20	-	mA	
LED Forward Voltage	V _{LED}	-	-	3.2	V	

3-3-2 LED Rank

Luminous Flux Rank Chart



Color Rank Chart



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3-4. LCD INTERFACE CONNECTIONS

Interface Connector: 20660-039E-01 (IPEX)

Table 3.4 LCD Connector Pin Configuration

Pin No.	Symbol	I/O	Description	Pin No.	Symbol	I/O	Description
1	VLED	P	LED+	21	GND	P	Ground
2	VLED	P	LED+	22	VDD3V3	P	Power Supply, 3.3V
3	NC		NC	23	VDD3V3	P	Power Supply, 3.3V
4	FB1	O	LED-	24	GND	P	Ground
5	FB2	O	LED-	25	D0N	I/O	MIPI Input Data Pair
6	FB3	O	LED-	26	D0P	I/O	MIPI Input Data Pair
7	NC		NC	27	GND	P	Ground
8	TP_RST	I	Signal for TP	28	D1N	I	MIPI Input Data Pair
9	TP_ATTN	O	Signal for TP	29	D1P	I	MIPI Input Data Pair
10	TP_SDA	I/O	Signal for TP	30	GND	P	Ground
11	TP_SCL	I	Signal for TP	31	CLKN	I	MIPI Input Clock Pair
12	GND	P	Ground	32	CLKP	I	MIPI Input Clock Pair
13	NC		NC	33	GND	P	Ground
14	VDDIN_TP	P	Power for TP	34	D2N	I	MIPI Input Data Pair
15	GND	P	Ground	35	D2P	I	MIPI Input Data Pair
16	RESET	I	Device Reset Signal	36	GND	P	Ground
17	LEDPWM	O	PWM output	37	D3N	I	MIPI Input Data Pair
18	NC		NC	38	D3P	I	MIPI Input Data Pair
19	ID1		ID1(Connector to 1.8 V)	39	GND	P	Ground
20	ID2		ID2(Connector to GND)				

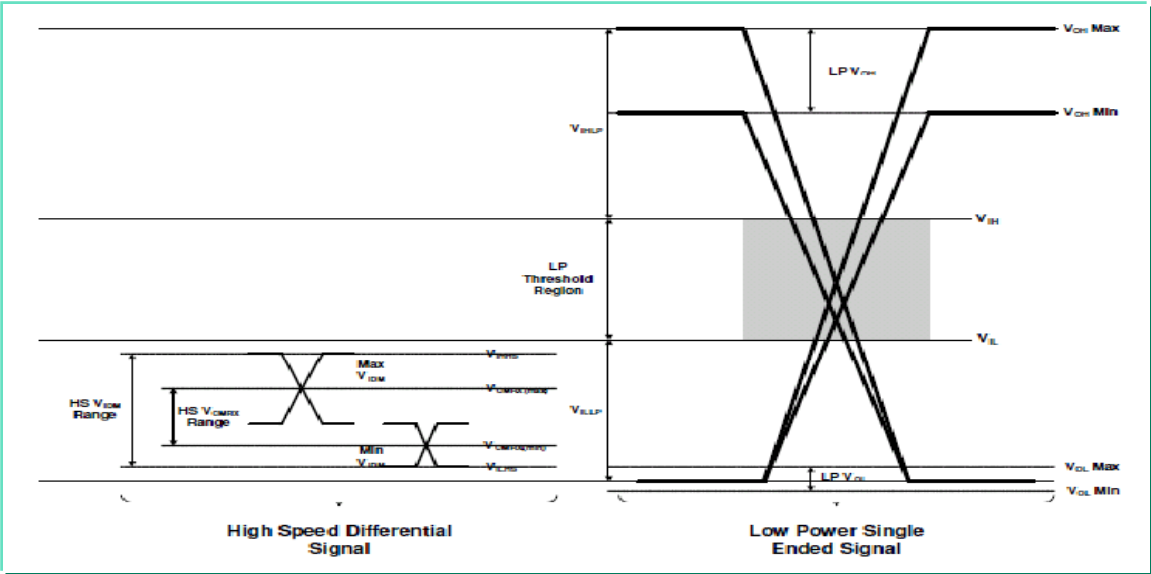
Table 3.5 TP Connector Pin Configuration

Pin No.	Symbol	I/O	Description	Pin No.	Symbol	I/O	Description
1	GND	P	Ground	6	VDDI_TP	P	Power Supply for TP
2	SDA	I/O	Serial data for TP	7	RST	I	Reset pin for TP
3	SCL	I	Serial CLOCK for TP	8	GND	P	Ground
4	GND	P	Ground	9	NC		NC
5	INT	O	interrupt pin for TP				

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3-5. SIGNAL TIMING SPECIFICATIONS

Parameter	Symbol	Min	Typ	Max	Unit	Condition
MIPI digital operation current	I _{VCCIF}	-	90	100	mA	带power IC
MIPI digital stand-by current	I _{VCCIFST}	-	10	15	mA	带power IC
MIPI Characteristics for High Speed Receiver						
Single-ended input low voltage	V _{ILHS}	-40	-	-		
Single-ended input high voltage	V _{IHHS}	-	-	460	mV	
Common-mode voltage	V _{CMRXDC}	70	-	330	mV	
Differential input impedance	Z _{ID}	80	100	125	Ω	
HS transmit differential voltage(V _{OD} =V _{DP} -V _{DN})	V _{OD}	70	-	250	mV	
MIPI Characteristics for Low Power Receiver						
Pad signal voltage range	V _I	-50	-	1350	mV	
Ground shift	V _{GNDSH}	-50	-	50	mV	
Output low level	V _{OL}	-50	-	50	mV	
Output high level	V _{OH}	1.1	1.2	1.3	V	

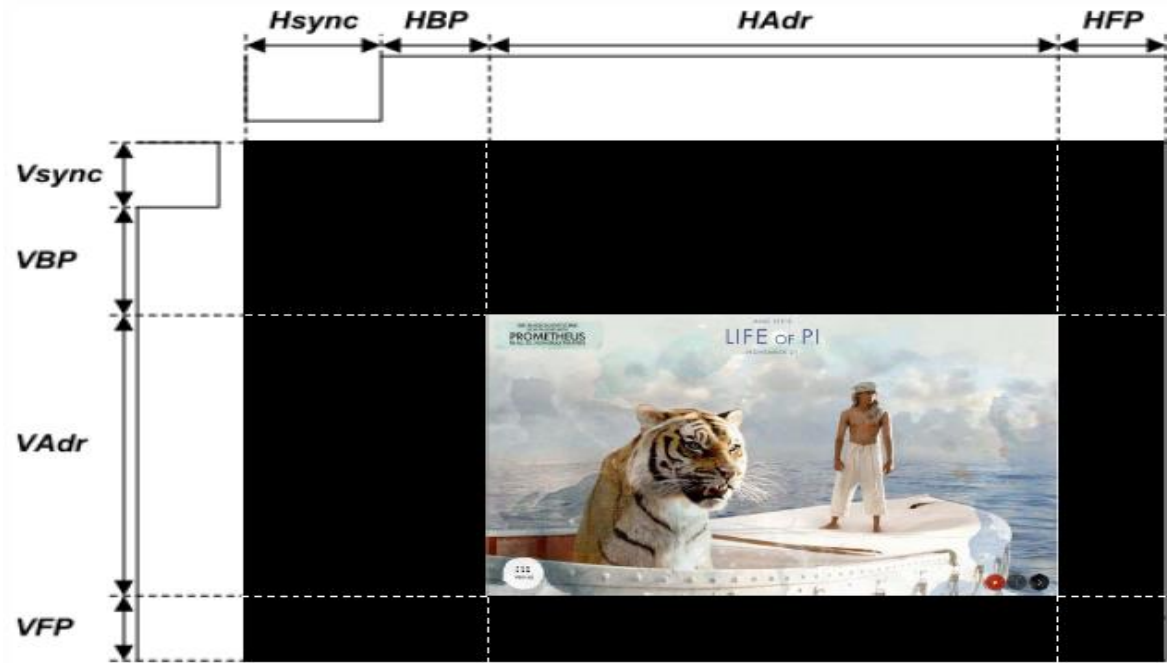


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3-5. SIGNAL TIMING SPECIFICATIONS

Lane数：3Lane

Item	Symbol	Min	Typ	Max	Unit
Pixel CLK	Tpixclk	-	40	-	MHz
MIPI CLK	Period	-	6.10	-	ns
	Frequency	-	164	-	MHz
Hsync	Period	-	4	-	t _{pCLK}
	Frequency	-	63.4	-	KHz
Vsync	Period	-	4	-	Line
	Frequency	-	60	-	Hz
Horizontal Active Display Term	HAdr	-	600	-	t _{pCLK}
	HBP	-	12	-	t _{pCLK}
	HFP	-	16	-	t _{pCLK}
	Total	-	632	-	t _{pCLK}
Vertical Active Display Term	Vadr	-	1024	-	Line
	VBP	-	12	-	Line
	VFP	-	16	-	Line
	Total	-	1056	-	Line



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3-6. MIPI Data&CLK Line Impedance Test Result

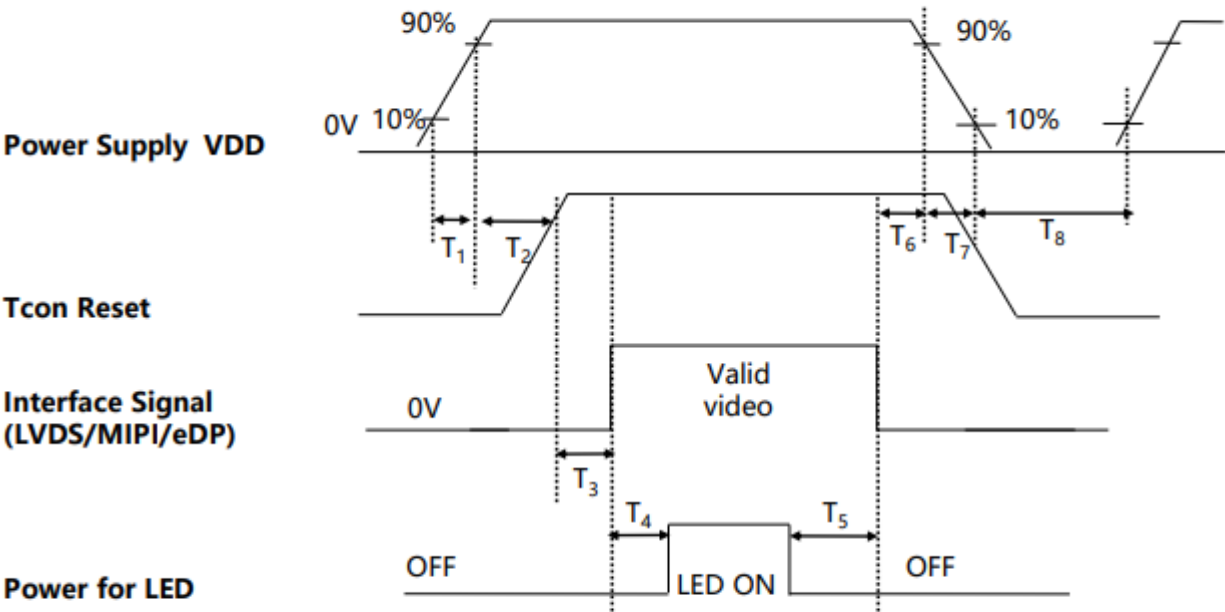
Item	1#	2#	3#	4#	5#
D3N/D3P	102	105	106	104	105
D2N/D2P	103	102	105	104	103
CLKN/CLKP	99	102	105	103	101
D1N/D1P	100	102	99	102	105
D0N/D0P	101	105	103	99	102

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3-7. Power On/Off Sequence

注：Reset 电压和VDD 电压需保持一致

[Ta =25±2 °C]



< Table15. Sequence Table >

Parameter	Value			Units
	Min.	Typ.	Max.	
T1	0.1	-	5	(ms)
T2	10	-	30	(ms)
T3	5	-	100	(ms)
T4	200	-	-	(ms)
T5	200	-	-	(ms)
T6	0	-	50	(ms)
T7	0	-	10	(ms)
T8	500	-	-	(ms)

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3-8. Software Flow

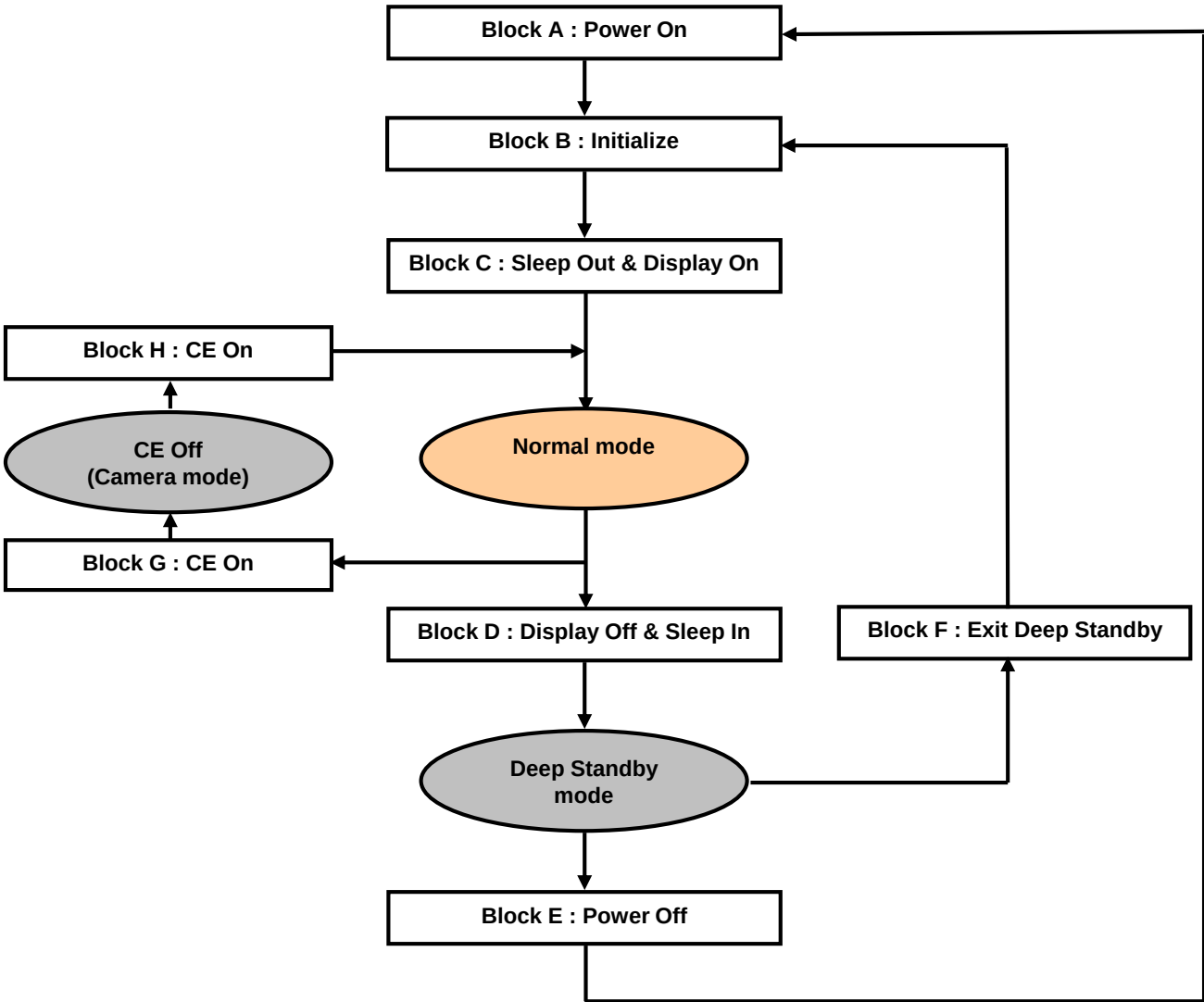


Fig 3.7 Software Flowchart

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3-9. Initial Code (3lane)

```
{  
{0x11,0,{0x00}},  
{REGFLAG_DELAY, 120, {0}},  
  
// Display ON  
{0x29,0,{0x00}},  
{REGFLAG_DELAY, 20, {0}},  
  
{0x35,1,{0x00}},
```

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3-10. IC General Spec and Size

2、Description

This document describes Chipone's ICNL9706AC-07 supports WXGA resolution driving controller. It includes a timing controller with glass interface level-shifters and a glass power supply circuit to drive a dot-matrix TFT LCD with 800 (RGB) x1600 dots at maximum.

The ICNL9706AC-07 supports MIPI DSI (Display Serial Interface) interface mode.

- Four data lanes support up to 550Mbps on the MIPI DSI.

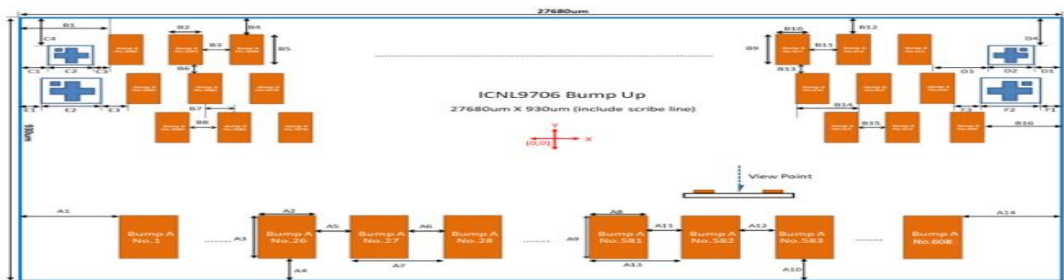
The ICNL9706AC-07 also supports various functions to reduce the power consumption of a LCD system via software control.

- Sleep In mode
- Deep standby mode

The ICNL9706AC-07 is suitable for any small portable battery-driven and long-term driving products, such as digital cellular phones, smart phones.

10.1、Chip Information

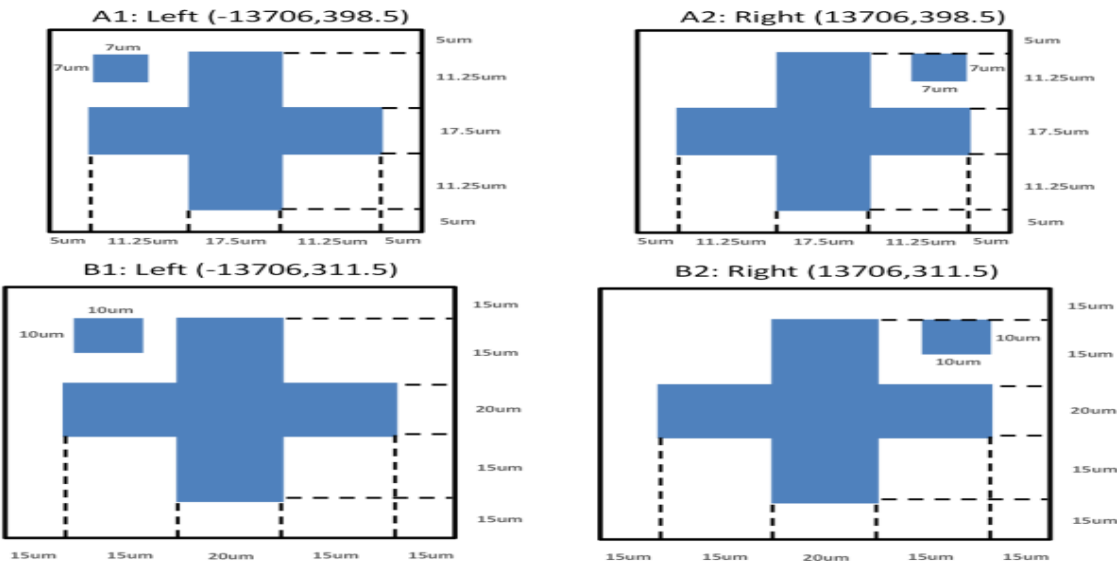
10.1.1 Pad Location



10.1.2 Chip Size

Chip size	X	Y	Unit
	27680	930	um

10.1.3 Alignment Mark



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3-11. Power Consumption

	Parameter	Symbol	Typ	Unit	Remarks
CABC off	Logic Power	L0	209.55	mW	
		L32	214.17	mW	
		L64	214.5	mW	
		L96	215.16	mW	
		L127	215.16	mW	
		L160	215.82	mW	
		L192	215.82	mW	
		L224	215.82	mW	
		L255	210.87	mW	
		R255	301.95	mW	
		G255	301.62	mW	
		B255	301.29	mW	
		8 color bar	280.5	mW	
		0-255 Gray Transition	215.82	mW	
		Flexible picture	245.19	mW	
	BLU	25% on	240.486	mW	
		50% on	498.9	mW	
		75% on	769.857	mW	
		100% on	1052.296	mW	

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4. OPTICAL CHARACTERISTICS

4-1. Optical Characteristics – Backlight 100%

Parameter	Symbol	Condition	Min	Typ	Max	Unit	Remarks
Viewing Angle	Θ12	CR >10	80	85	-	°	Note 1
	Θ6		80	85	-	°	Note 1
	Θ9		80	85	-	°	Note 1
	Θ3		80	85	-	°	Note 1
	Θ1	CR>10	80	85	-	°	Note 1
	Θ4		80	85	-	°	Note 1
	Θ7		80	85	-	°	Note 1
	Θ11		80	85	-	°	Note 1
Contrast Ratio	CR	Optimal	600	800	-	-	Note 1,4
Brightness	Lv	Optimal	300	350	-	Cd/m ²	Note 1
Brightness Uniformity	Y	Optimal	-	85	-	-	Note 1,7(5P)
	Y	optimal	75	80	-	-	Note 1.7(13P)
Flicker			-	-25	-	dB	Note 1,2
Crosstalk			-	2.2	-	%	Note 1,3
Response time	τ _f or τ _r	Θ =0 ° Ta =25 °C	-	30	40	ms	Note 1,6
Color Gamut	NTSC	-	45	50	-	%	Note 1
White Chromaticity	x	CIE 1931	0.265	0.295	0.325	-	Note 1
	y		0.285	0.315	0.345	-	Note 1
Red Chromaticity	x	CIE 1931	0.558	0.588	0.618	-	Note 1
	y		0.309	0.339	0.369	-	Note 1
Green Chromaticity	x	CIE 1931	0.275	0.305	0.335	-	Note 1
	y		0.545	0.575	0.605	-	Note 1
Blue Chromaticity	x	CIE 1931	0.122	0.152	0.182	-	Note 1
	y		0.091	0.121	0.151	-	Note 1

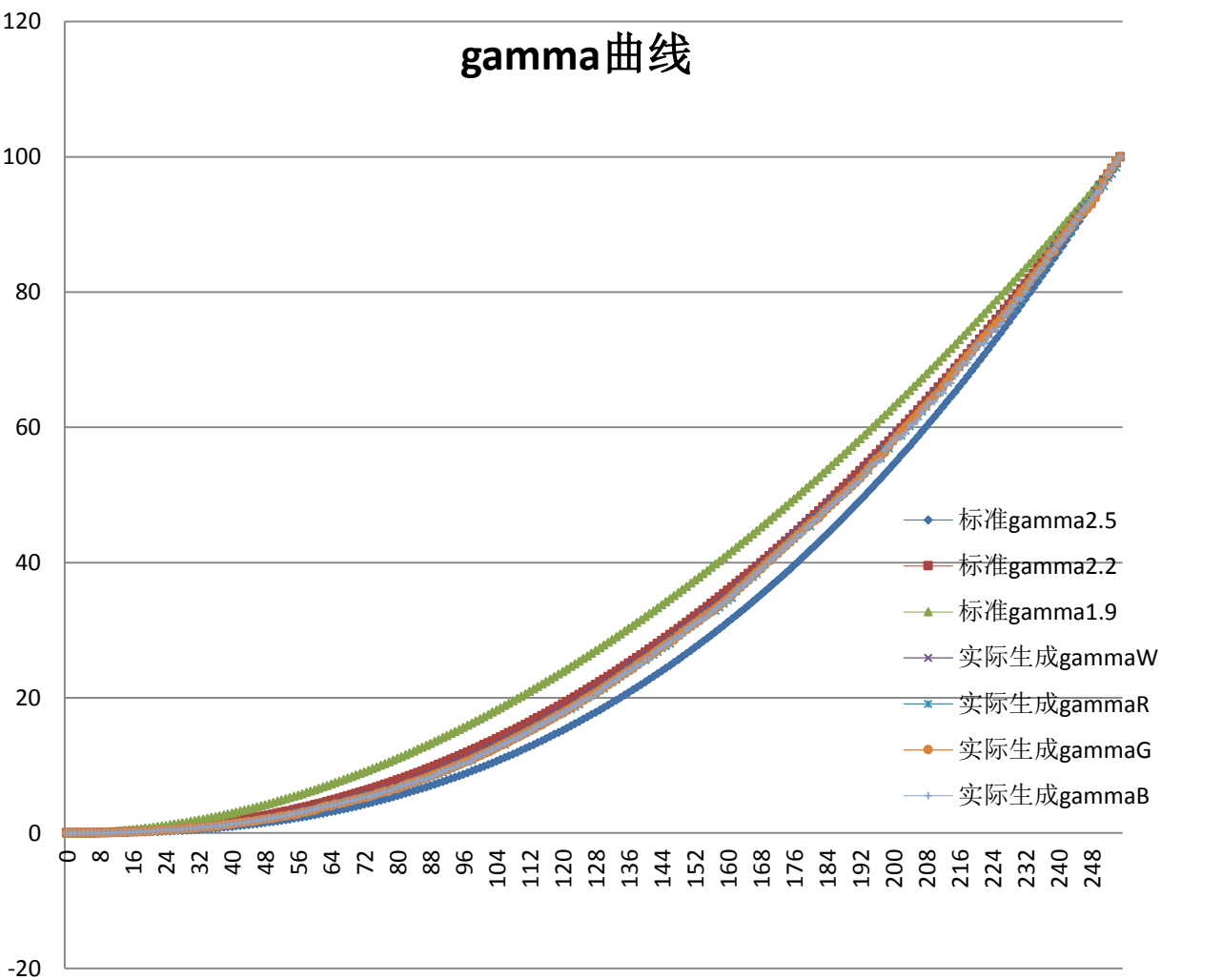
4-2. Cell&BLU Optical Characteristics

Parameter	Typ	Unit	Remarks
Aperture Ratio	70	%	
Upper Pol Trans.	43	%	
Lower Pol Trans.	43	%	
Panel Trans.	4.9(TYP)	%	w/o APF
Panel Trans.	-	%	with APF
BLU Luminance	8000	Cd/m ²	Center
BLU Luminance Uniformity	80	%	Note1,7

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4-3. Gamma/Color Coordinate Uniformity/CE Function/CABC Function

4.3-1 Gamma Curve
Request: R/G/B/W, 0-255 gray scale, step 1 gray scale




WRGB GAMMA
测试数据

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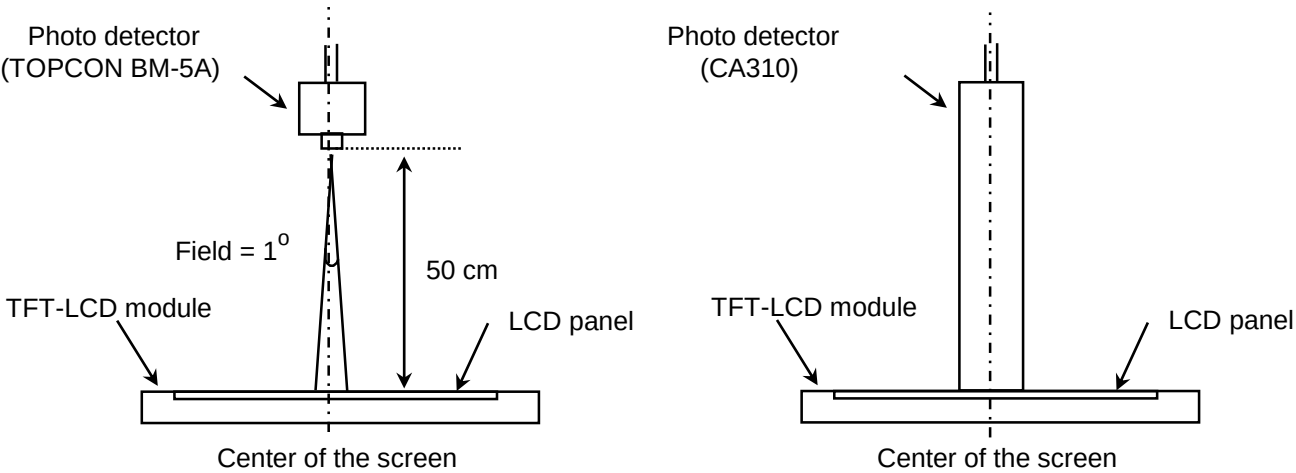
4-3. Gamma/Color Coordinate Uniformity/CE Function/CABC Function

4.3-2 Color Coordinate Uniformity
Request: white pattern, 0-255 gray scale, step 15 gray scale.

Gray Scale	x	y	White	x	y
0	-	-	135	0.288	0.310
15	0.283	0.312	150	0.288	0.310
30	0.288	0.317	165	0.288	0.310
45	0.288	0.317	180	0.288	0.310
60	0.288	0.315	195	0.288	0.310
75	0.287	0.313	210	0.287	0.310
90	0.287	0.312	225	0.287	0.310
105	0.287	0.311	240	0.287	0.312
120	0.288	0.310	255	0.288	0.312

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[Note 1] Optical Test Equipment Setup

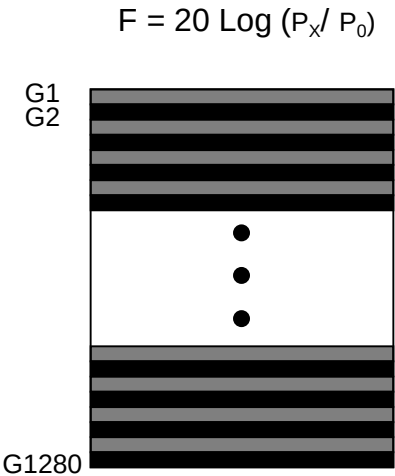


View angel range measurement setup Luminance , uniformity and color measurement setup

Fig 4.1. Optical Characteristic Measurement Equipment and Method

[Note 2] Flicker

The flicker level should be measured with horizontal gray/black stripes. The flicker is essentially a ratio of the powers in the frequency spectrum at 30 Hz (P_x) and 0 Hz (P_0 , DC level).



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[Note 3] Crosstalk

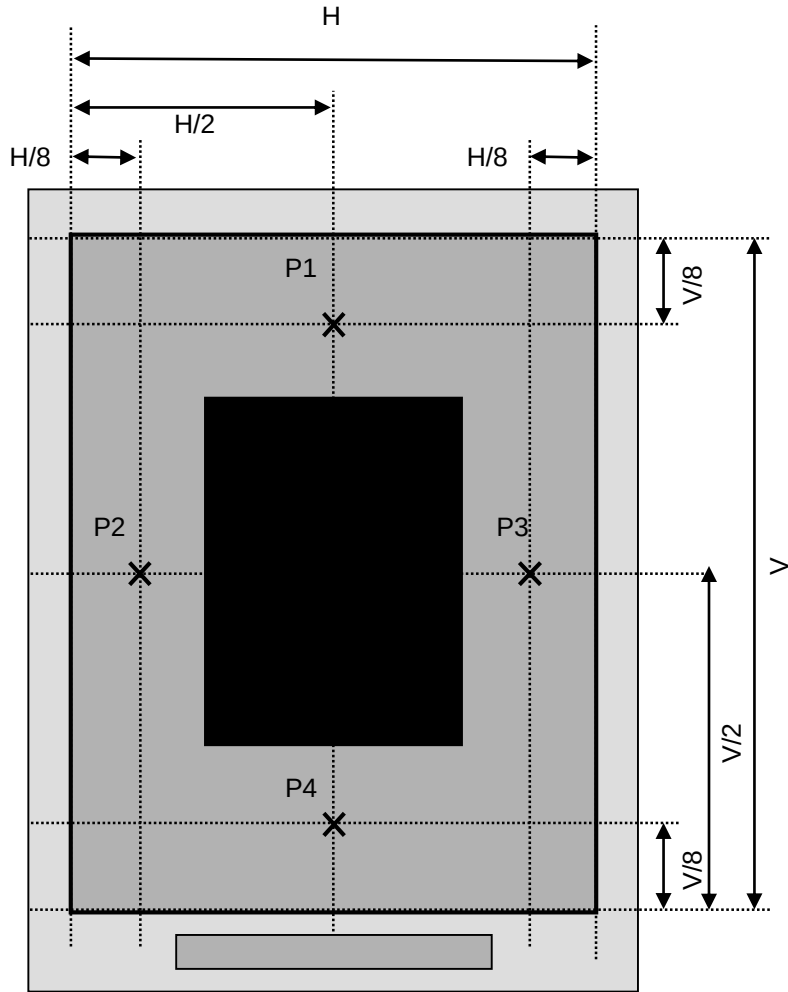


Fig 2-5. Crosstalk measurement points

A: Luminance for P1 ~ P4 with all 127gray pixels

B: Luminance for P1 ~ P4 with 127gray pixels when the black box is applied

$$\text{Crosstalk [\%]} = \text{Maximum} \left[\text{Absolute} \left(\frac{A - B}{A} \right) \right]$$

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[Note 4]

Contrast Ratio is defined as follows ;

Contrast Ratio(CR) =
$$\frac{\text{Photo detector output with LCD being "White"}}{\text{Photo detector output with LCD being "Black"}}$$

[Note 5]

Viewing Angle Range is defined as follows;

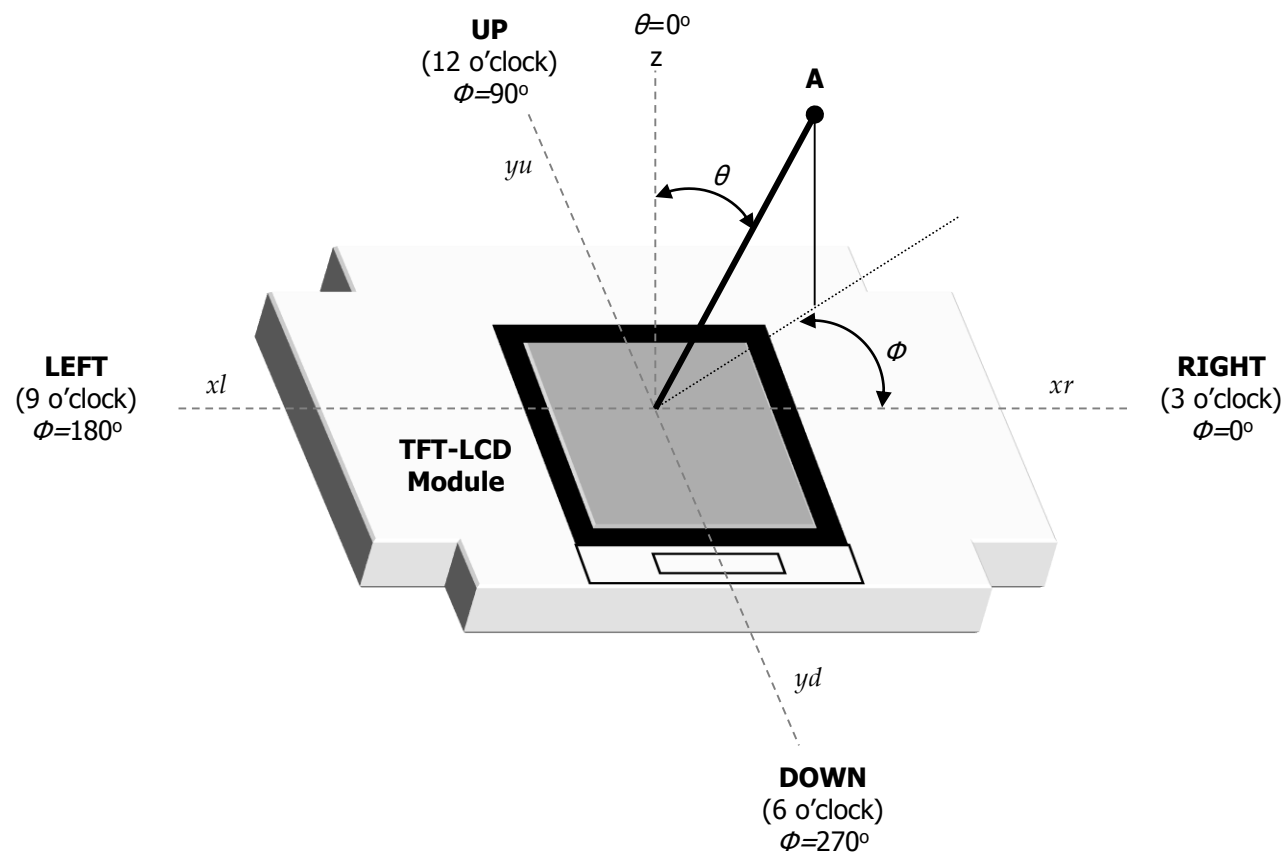
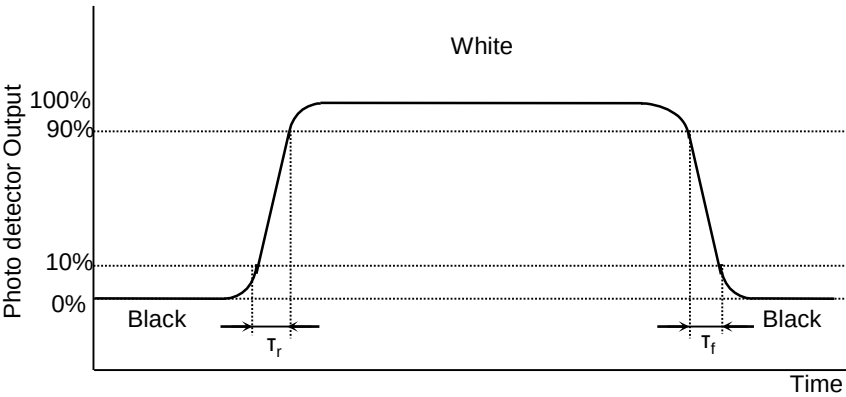


Fig 4.2 Viewing Angle Definitions

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[Note 6]

Response time is obtained by measuring the transition time of photo detector output, when input signals are applied so as to make the area “black” to and from “white”.

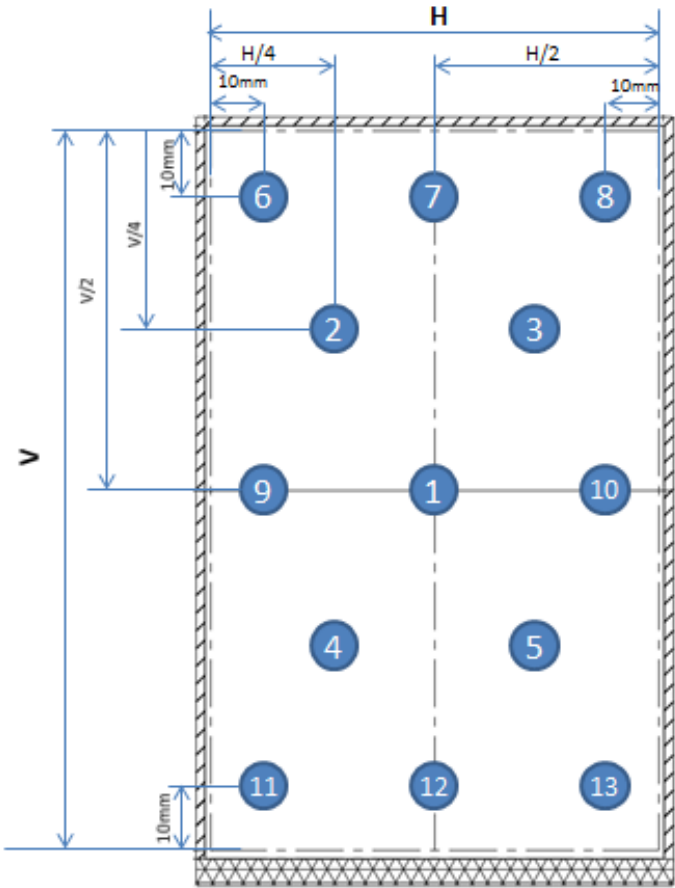


[Note 7]

Fig 4.3 Response Time Definition

The brightness measurement is taken at point 5P/13P.

Brightness Uniformity = $\frac{\text{Minimum Photo detector output for P1-P5(P13) with all pixels white}}{\text{Maximum Photo detector output for P1-P5(P13) with all pixels white}} \times 100$



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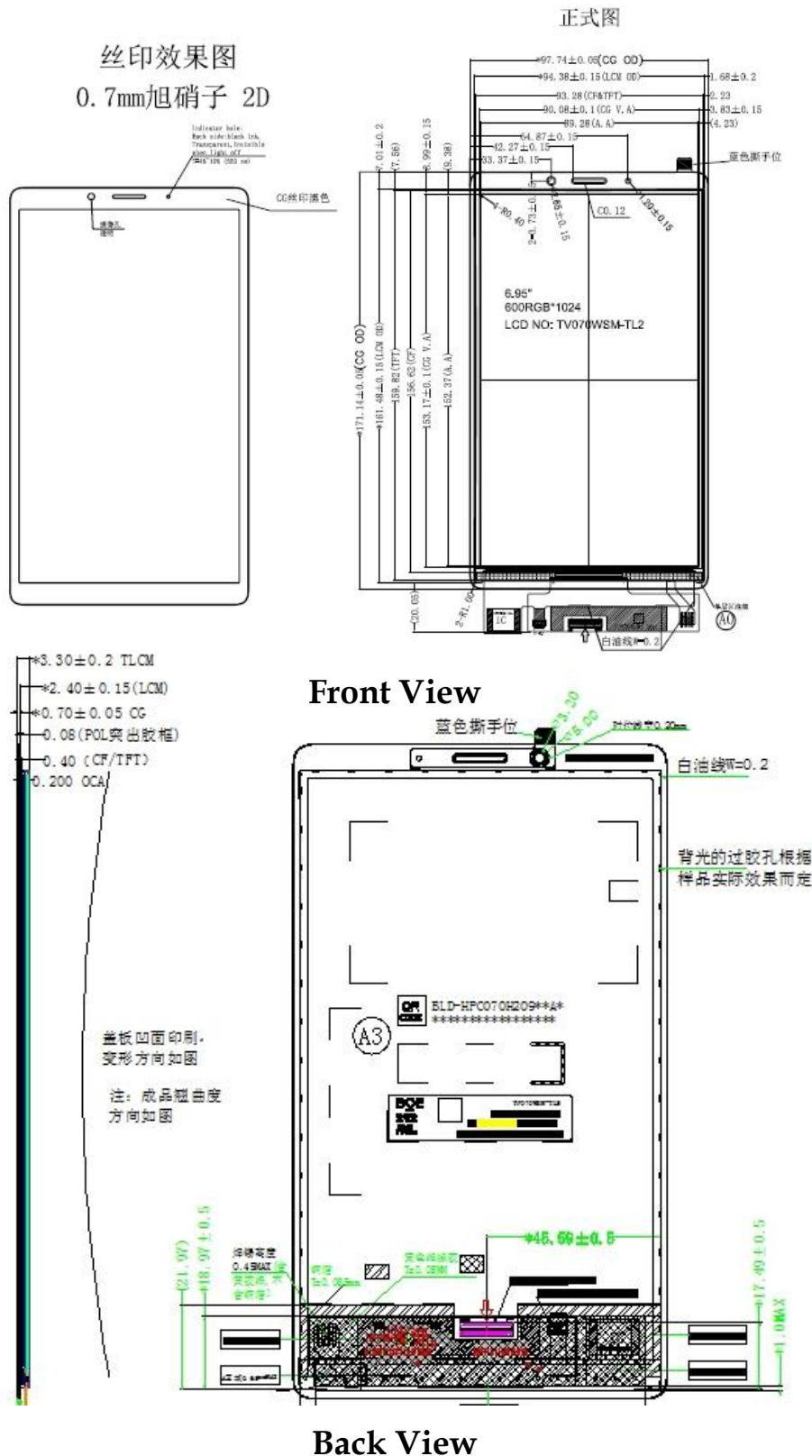
5. MECHANICAL CHRACTERISTICS

The contents provide general mechanical characteristics for the model.
In addition the figures in the next page are detailed mechanical drawing of the LCD.

Items	Dimension	Typ.	Tolera nce	Unit
Mother Glass	Size	1500*1850	-	mm
Q-Panel	Size	Q1 : 476.40×327.64mm Q2 : 476.40×495.46mm	-	mm
C/F and TFT thickness	thickness	0.4+0.4		mm
Panel	A/A	89.28*152.3712		mm
	C/F	93.28*156.62		mm
	TFT	93.28*159.82		mm
	BM(U/D/L/R)	1.8288/2.42/2.0/2.0	-	mm
	IC Bonding Area	0.85		mm
	Pol Size	154.85*92.68mm	±0.15	mm
	Gap Between Pol~Glass(U/D/L/R)	0.3/0.3/0.3/4.67	±0.15	mm
Module	Horizontal	97.74	±0.05	mm
	Vertical	171.14	±0.05	mm
	Thickness	3.3	±0.2	mm
	UV Glue Thickness	-	-	mm

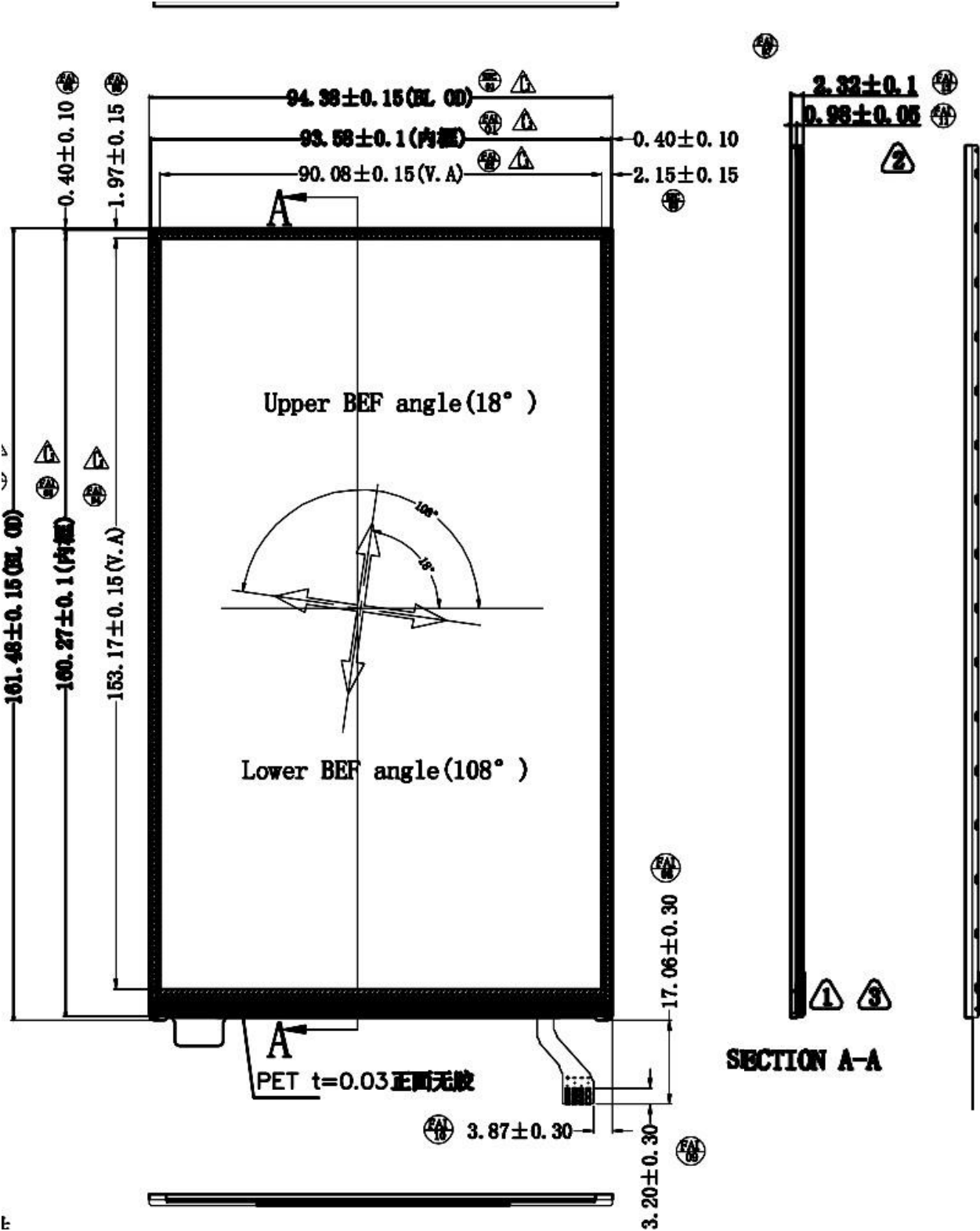
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5.1 LCM Drawing (With TP)
Folded and unfolded status



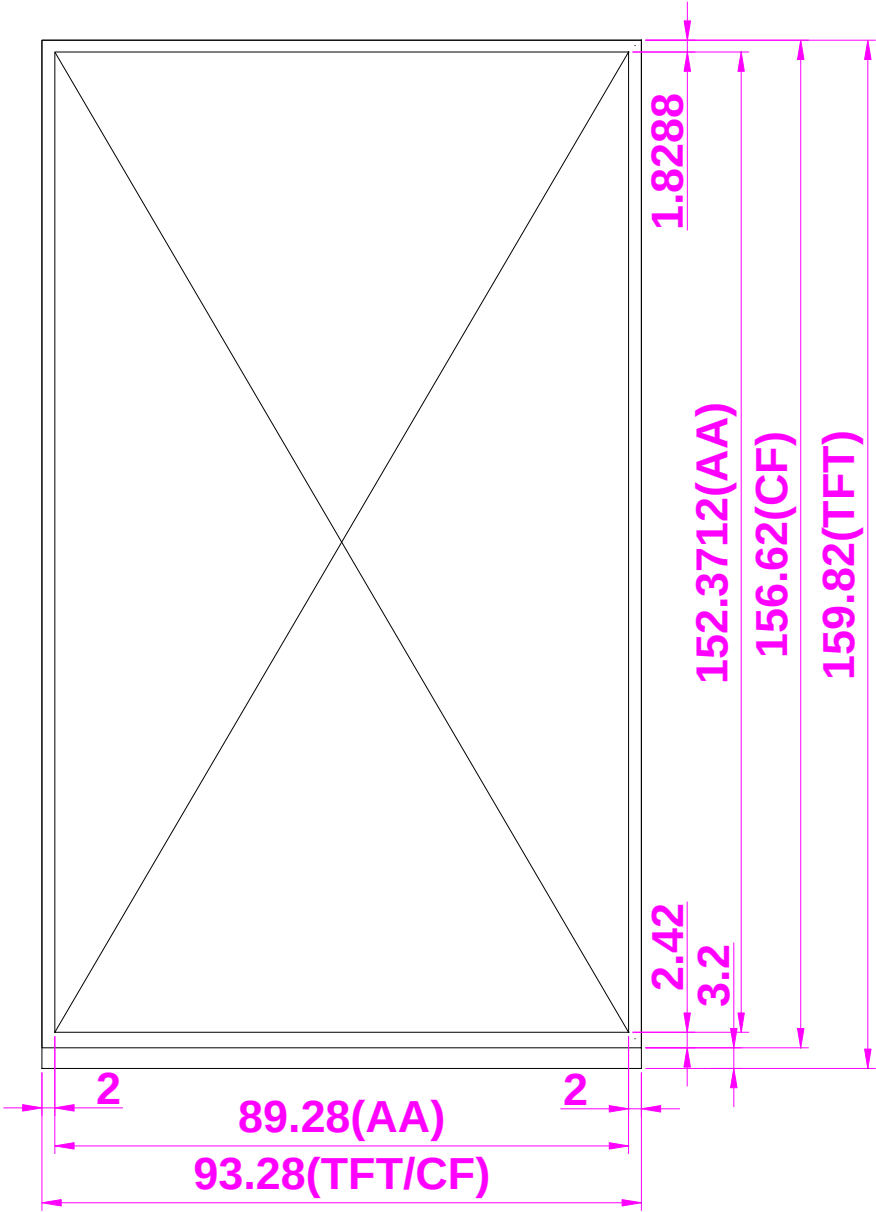
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5.2 BLU Outline Dimension
Including each film



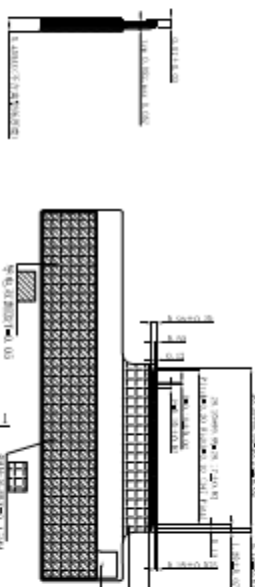
Product Specification

5.3 Panel Outline Dimension



5.4 FPCA Outline Dimension

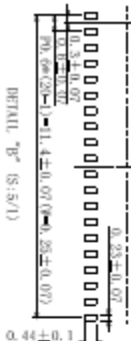
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REV.	DESCRIPTION	REVISER	DATE
AD	NEW (GH209 5000 1.50D PIN)	WLL	2020-04-01

PIN DESCRIPTION

TP P/N 注文	Signal	TP P/N 注文	Signal
PTV	Motor	PTV	Motor
1	2800	1	VE130
2	S04	2	VE130
3	S02	3	NC
4	000	4	LE101
5	187	5	NC
6	VE100	6	LE102
7	857	7	NC
8	030	8	LE003
9	NC	9	NC



Pin	Signal	Pin	Signal
1	TP PIN 定义	13	TP PIN 定义



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respected distance:	40.10	UNITS: mm	DATE:	PART NUMBER:
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DESIGN BY:	WLL	2020.04.01
DESIGNED BY:	LHR	2020.04.01

FWC-HPIC070H250-M0/T

SHEET:	1 OF 1
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DATE	20.04.01	DO NOT SCALE THIS DRAWING
Autodesk		

PROJECTIONS	
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DRAWING DESCRIPTION:	MODEL
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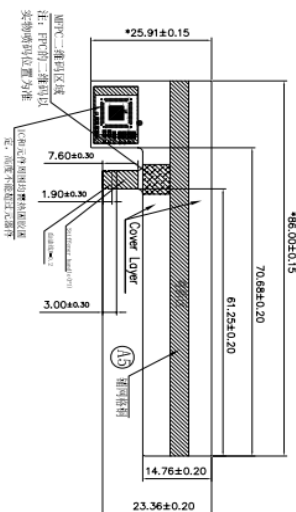
[illegible][illegible]

製品制作

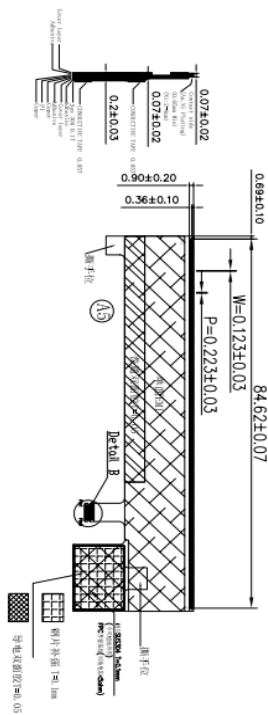
5.5 TFPCA Outline Dimension

注意：所有走线都需要倒弧度处理。

正视图



侧视图



背视图

MARK	REV.	DESCRIPTION	REVISER	DATE
	A0	NEW (双面胶增加绑手位)	WLL	2019.03.22
	A1	双面胶增加绑手位	WLL	2019.03.27
⑫	A2	更改双面胶绑手位位置	WLL	2019.05.08
⑬	A3	改善通双面胶为导电布, 同步更改外形	WLL	2019.05.30
⑭	A4	导电布外形加大	WLL	2019.06.12
⑮	A5	增加单面DRL, 导电布改为双面胶, 外形减小, 单 层区改为两层结构	WLL	2019.06.25
	A5	增加二维码	WLL	2019.07.05

Pin	Signal Name	Pin	Signal Name
1	GND	1	VLED+
2	SDA	2	VLED+
3	SCL	3	NC
4	GND	4	LED1
5	INT	5	NC
6	VDDIO_X_TP	6	LED2
7	RST	7	NC
8	GND	8	LED3
9	NC	9	NC

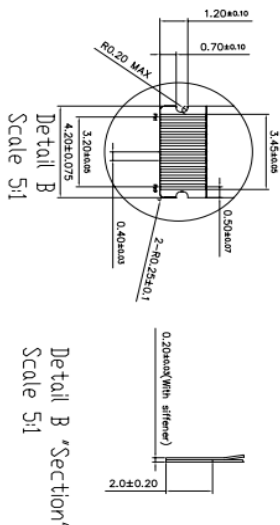
1.COMPONENTS HEIGHT 1.0mm(Max.)
FPCA HEIGHT 1.35mm(Max.)

2. GROUNDING AREA:

3. Halogen Free & RoHS & Reach

注: IC由我司提供, 其余供应商自备。同种类别的元件只能选同一个产家。

Stack-up Totals		
NO	Brand	Resistor (m)
1	AI	0.00-0.2
2	W	2-4
2	EM	-
3	CONLEY	12.5
4	Adaptive	15
5	Ca	12-48
6	W	25
7	Ch	17-48
8	Adaptive	15
9	CONLEY	12.5
10	W	-
11	EM	-
12	Adaptive	40
13	5050A	100
14	CONLEY W	80



NOTES:

1. 刀模公差: $\pm 0.2\text{mm}$, 钢板公差: $\pm 0.1\text{mm}$
2. 未注公差: 外形 $\pm 0.15\text{mm}$, 压铸公差: $\pm 0.05\text{mm}$
3. 料强平坦度要求通过高度不超过材料对角长度 0.75%
4. 所有材料符合 RoHS2.0 & HALOGEN FREE & REACH
5. 标注 * 为关键点, 未标注尺寸以实际为准
6. 金手指尺寸测量以锡层根部为准
7. FPC 材质: 无卤电解铜
8. FPC 来料用 Tray 盘包装, 尺寸不大于长 * 宽 * 高: 460×350 .



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Autodesk	UNSPECIFIED TO ORIGIN:	±0.20	(UNITS: mm)	DATE:	PART NUMBER : PFC-HPC070H209-01-A5		PROJECTION	DO NOT SCALE THIS DRAWING.	DESCRIPTION:	SHEET:	MODEL
	DISTION BY:	WLL	2019, 06, 25	1 OF 1							
	CHECKED BY:		19, 06, 25								

Product Specification

6. RELIABILITY TEST

Must be accordance with Lenovo RA test items

测试项目	测试条件	样品形态	样品数量	评价结果	Remark
-	初检	-	MDL	L0漏光	0.4t glass验证@DV4
光学	Luminance	300MIN/350TYP	TLCM	20pcs	450-500nit
	uniformity	75%MIN/80%TYP	TLCM	5PCS	80%-81%
	Contrast	800:1 TYP	TLCM	20pcs	1100-1400
	NTSC	45%MIN/50%typ	TLCM	20pcs	48%-50%
	Flicker	-30dB	TLCM	20pcs	-33dB—36dB
	Image sticking	1S	TLCM	5PCS	1S
电学	功耗测定	NA	LCD	10PCS	Iwr:47mA-50mA, Isleep:8mA
	功耗测定	60mA	BLU	10PCS	59.7-59.8mA
RA环境	HTO	60°供电、240H	TLCM	5PCS	PASS
	LTO	-20°供电、240H	TLCM	5PCS	PASS
	HTBO	60°/90° 供电、240H	TLCM	5PCS	PASS
	HTS	80°储存、240H	TLCM	5PCS	PASS
	LTS	-40°储存、240H	TLCM	5PCS	PASS
	HTBS	60°/90° 储存、240H	TLCM	5PCS	PASS
	Thermal shock test	-40°C→60°C/ 30cycle	TLCM	5PCS	PASS
机械环境	Panel 4轴 bending (TFT)	ε量信下限≥1500μ	Cell	30PCS	1886μ
	Panel 4轴 bending (CF)	ε量信下限≥1500μ	Cell	30PCS	1761μ
	FOG拉力强度	≥7N/cm	FOG	20pcs	8.8N-12.4N
	IC 3轴 bending	B10≥350Mpa	IC	20pcs	381Mpa
	小球跌落试验	直径Φ25mm, weight 130g,防静电皮厚度 2mm,16CM	TLCM	5PCS	pass

Product Specification

7. Package

7.1. Packing Description

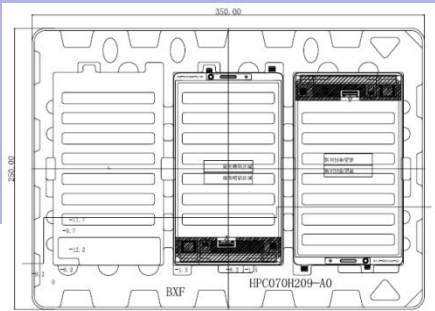
- 1. Each big box contains four small boxes:4 boxe
- The stacked tray per a small box : 9pcs tray
- Full (LCD Included) tray 32pcs + Empty tray 4pcs
- 2. Stacking Method of Trays : Zigzag

-. Cover 表面贴附保护膜后

将 3pcs MDL 平放入Tray

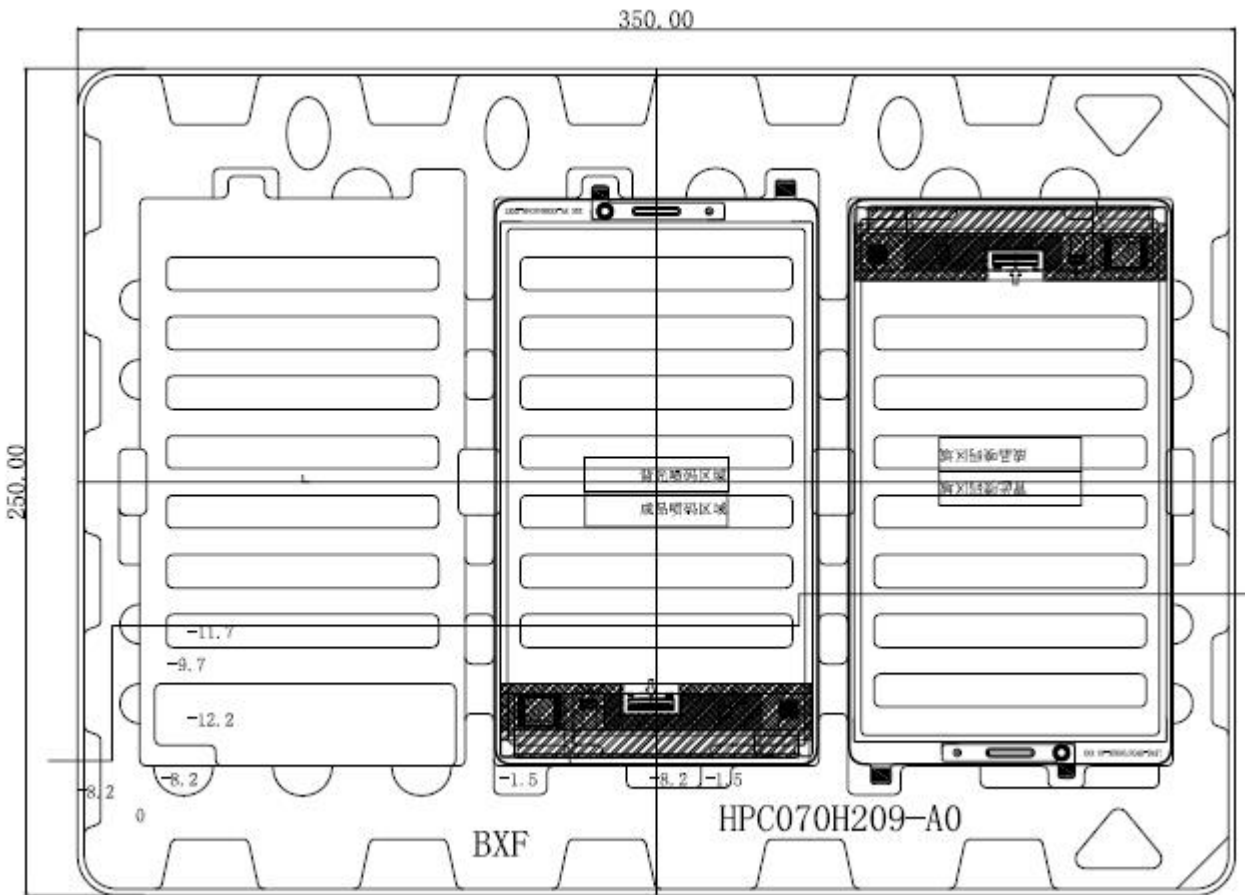
-. 将8pcs PET Tray 平放入Anti-static vacuum bag;

-. Tray 无需旋转放置, 顶部1pcs 空Tray;



Product Specification

7.2. Description of Packing Tray



Product Specification

7.3. Description of Packing Box

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一、包装式样:

1. 将完成线路板的产品收入包装袋内
2. 将包装袋放入产品盒内，盒盖一合，将产品盒放入外包装箱内，盖上泡沫垫，放入一个空包装袋
3. 将泡沫垫一拆，将产品放入外包装箱内，放入一包干燥剂，放入一包干燥剂
4. 将外包装箱一拆，将产品放入外包装箱内

二、出货标签:

Label Size: 115*455

白色字体为打印标识，说明如下:

1. 产品型号: TV070WSM-TL5
2. 产品数量
3. 日期: 2020-04-27
4. 产品日期: 2020-04-27
5. 产品日期: 2020-04-27
6. 产品日期: 2020-04-27
7. 产品日期: 2020-04-27

三、外箱出货标识:

空白标签，所有字体均为打印，说明如下:

1. 产品型号: TV070WSM-TL5
2. 产品数量
3. 日期: 2020-04-27
4. 产品日期: 2020-04-27
5. 产品日期: 2020-04-27
6. 产品日期: 2020-04-27
7. 产品日期: 2020-04-27

备注: 所有材料符合RoHS 2.0

材料	规格	数量	单位
外包装	200x200x200 (mm)	1	个
内盒	150x150x150 (mm)	4	个
泡沫垫	100x100x100 (mm)	4	个
干燥剂	100x100x100 (mm)	4	个
包装袋	100x100x100 (mm)	4	个

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Product Specification**8.0 Handling & Cautions****(1) Cautions when taking out the module**

- Pick the pouch only, when taking out module from a shipping package.

(2) Cautions for handling the module

- As the electrostatic discharges may break the LCD module, handle the LCD module with care. Peel a protection sheet off from the LCD panel surface as slowly as possible.
- As the LCD panel and back - light element are made from fragile glass material, impulse and pressure to the LCD module should be avoided.
- As the surface of the polarizer is very soft and easily scratched, use a soft dry cloth without chemicals for cleaning.
- Do not pull the interface connector in or out while the LCD module is operating.
- Put the module display side down on a flat horizontal plane.
- Handle connectors and cables with care.

(3) Cautions for the operation

- When the module is operating, do not lose CLK, ENAB signals. If any one of these signals is lost, the LCD panel would be damaged.
- Obey the supply voltage sequence. If wrong sequence is applied, the module would be damaged.

(4) Cautions for the atmosphere

- Dew drop atmosphere should be avoided.
- Do not store and/or operate the LCD module in a high temperature and/or humidity atmosphere. Storage in an electro-conductive polymer packing pouch and under relatively low temperature atmosphere is recommended.

(5) Cautions for the module characteristics

- Do not apply fixed pattern data signal to the LCD module at product aging.
- Applying fixed pattern for a long time may cause image sticking.

(6) Other cautions

- Do not disassemble and/or re-assemble LCD module.
- Do not re-adjust variable resistor or switch etc.
- When returning the module for repair or etc., Please pack the module not to be broken. We recommend to use the original shipping packages.